

Netflix Data Analysis Report

Project Title

Netflix Data Analysis Using Python

Introduction

Netflix is one of the world's leading online streaming platforms that provides a wide variety of movies, TV shows, documentaries, and entertainment content across different countries and genres. With the rapid growth of digital streaming services, analyzing Netflix data helps in understanding audience preferences, content trends, and global distribution patterns.

This project focuses on analyzing the Netflix dataset using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn. The analysis provides meaningful insights into content growth, genre popularity, rating distribution, and country-wise contribution on Netflix.

The project also demonstrates how data analytics and visualization techniques can be used to extract valuable business insights from large datasets.

Objectives

The main objectives of this project are:

- To analyze Netflix movies and TV shows dataset.
 - To perform data cleaning and preprocessing.
 - To identify top countries producing Netflix content.
 - To analyze yearly trends of Netflix releases.
 - To compare Movies and TV Shows distribution.
 - To visualize genre and rating distributions.
 - To generate meaningful insights using data visualization techniques.
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Tools and Technologies Used

The following tools and technologies were used during the project:

- Python
- Pandas
- NumPy

- Matplotlib
 - Seaborn
 - Jupyter Notebook
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Dataset Information

The dataset contains detailed information related to Netflix content, including:

- Show ID
- Content Type (Movie / TV Show)
- Title
- Director
- Cast
- Country
- Date Added
- Release Year
- Rating
- Duration
- Genre

This dataset was used to perform data cleaning, exploratory data analysis, and visualization.

Data Cleaning and Preprocessing

Before analysis, several preprocessing steps were performed to improve data quality and ensure accurate results.

Steps Performed:

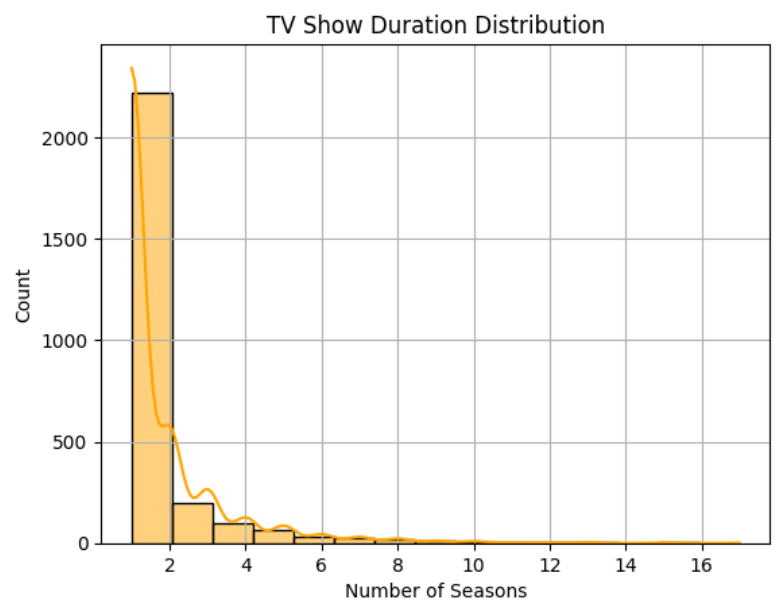
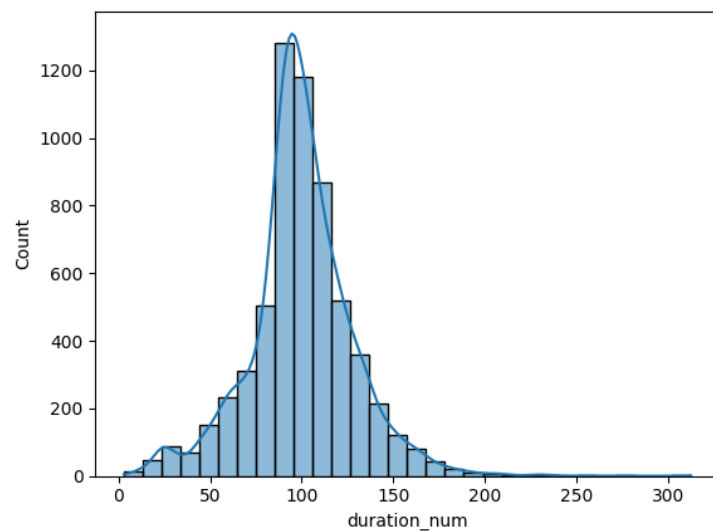
1. Checked dataset structure and column information.
2. Identified missing and null values.
3. Removed unnecessary or duplicate records.
4. Filled missing values in important columns:
 - Director → "Unknown"
 - Cast → "Unknown"
 - Country → "India"
5. Converted the date_added column into datetime format.
6. Created a new column named year_added for yearly trend analysis.

7. Verified data types and corrected formatting issues.

Proper preprocessing helped improve the accuracy and readability of the analysis.

Exploratory Data Analysis

1. Movie Durations & TV Show Seasons



Avg Movie Length

Minutes — peak at 100 mins; 150+ min films are typically documentaries or international

Typical TV Seasons

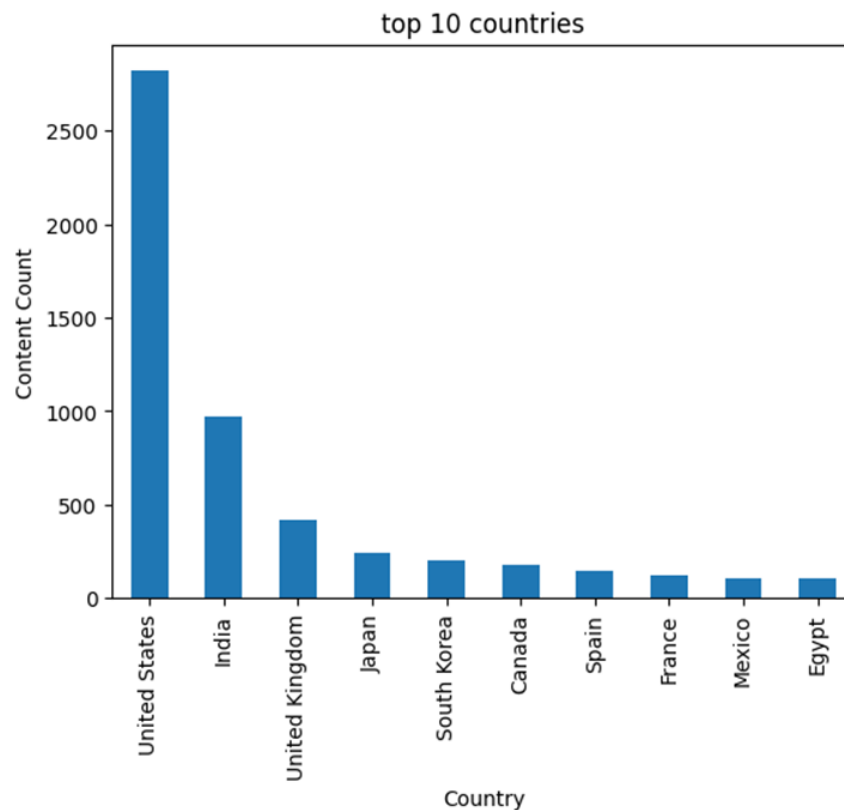
Most shows have 1–3 seasons, indicating high cancellation rates

Single-Season Shows

Of all TV content on the platform is single-season

2. Top Countries on Netflix

A bar chart was created to identify the countries producing the highest amount of Netflix content.

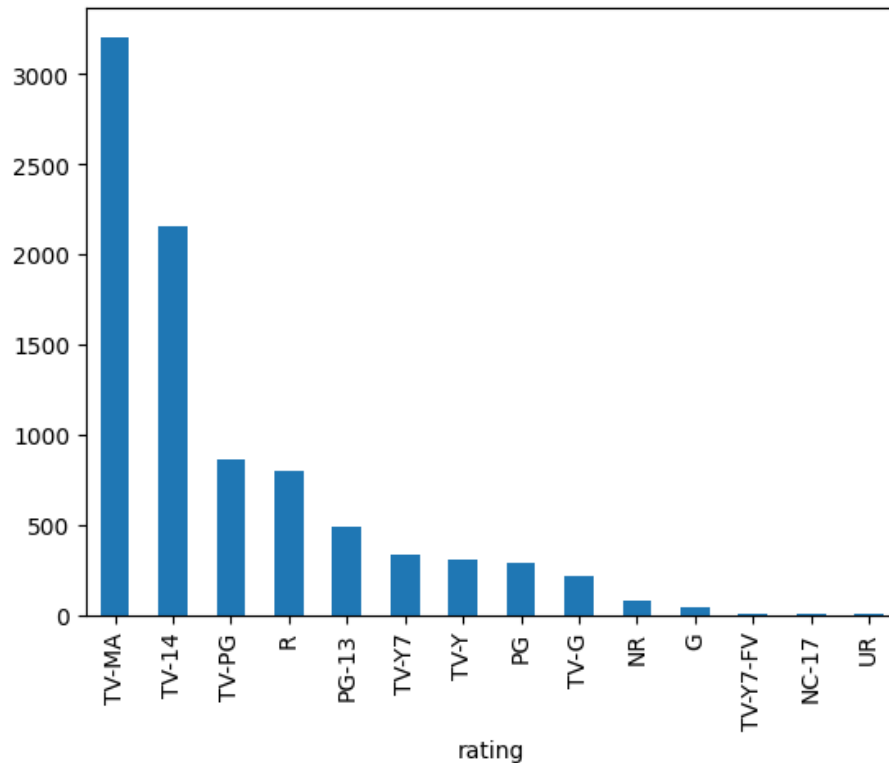


Observation:

- United States dominates the Netflix library with approximately 35% of total content.
- India ranks second with strong regional content growth.
- United Kingdom, Canada, and Japan are also major contributors.
- Emerging markets are steadily increasing their contribution year after year.
- Netflix has a strong international content distribution network.

3. Rating Distribution Analysis

A bar chart was created to analyze the distribution of Netflix content ratings across different audience categories.



Observation:

- TV-MA has the highest count among all ratings.
→ Most content on Netflix is made for mature audiences.
- TV-14 is the second most common rating category.
→ A large amount of content is suitable for teenagers above 14 years.
- TV-PG and R categories also contain a significant number of titles.
→ Netflix offers a balanced mix of family-friendly and adult-oriented content.
- Ratings like PG-13, TV-Y7, and TV-Y have comparatively fewer titles.
→ There is less children-focused content compared to adult content.
- Very few titles fall under NC-17, UR, and TV-Y7-FV categories.
→ Highly restricted or uncommon ratings are rare on the platform.

Overall Analysis:

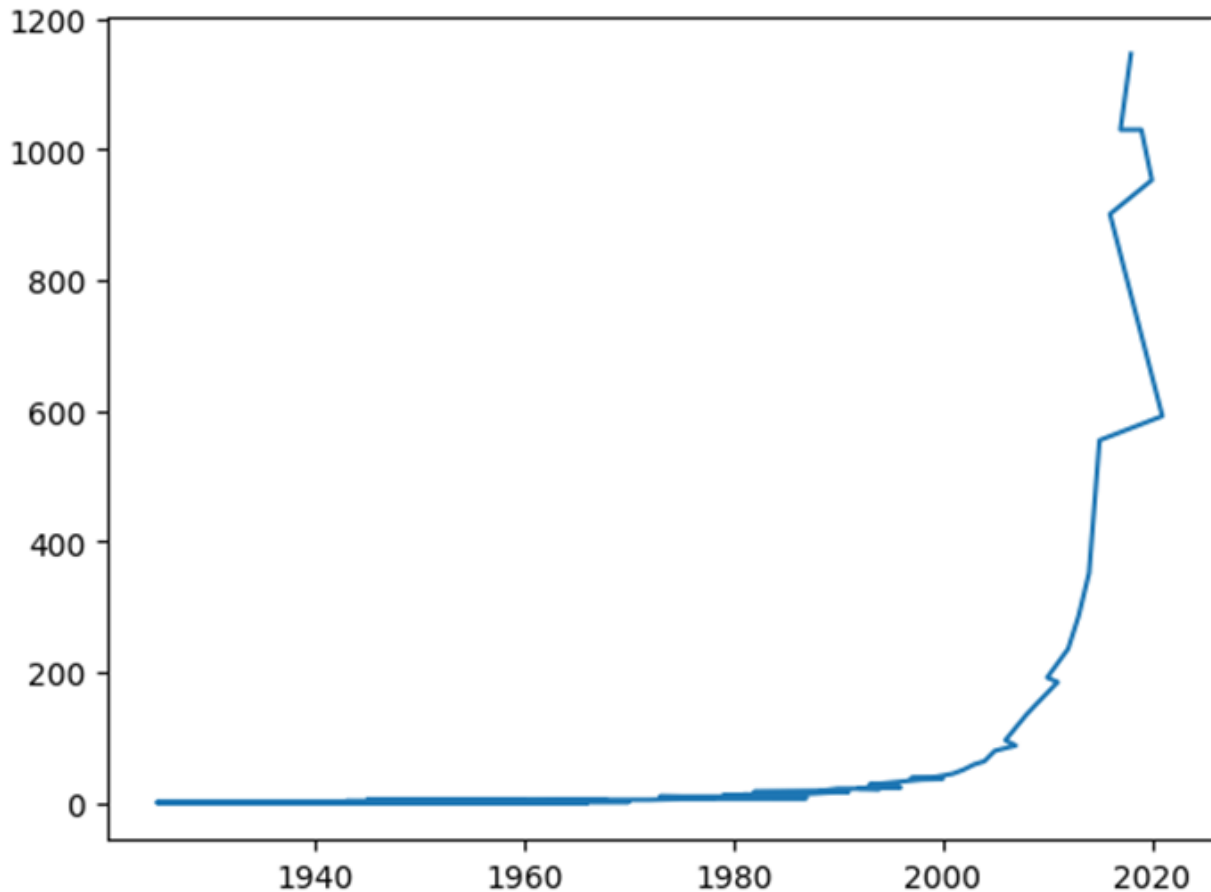
The platform mainly focuses on:

- Mature audience content
- Teen-oriented entertainment
- Limited extreme-restriction content

The rating distribution is right-skewed, where a few rating categories dominate the majority of Netflix titles.

4. Release Year Analysis

A line chart was used to analyze yearly content release trends.

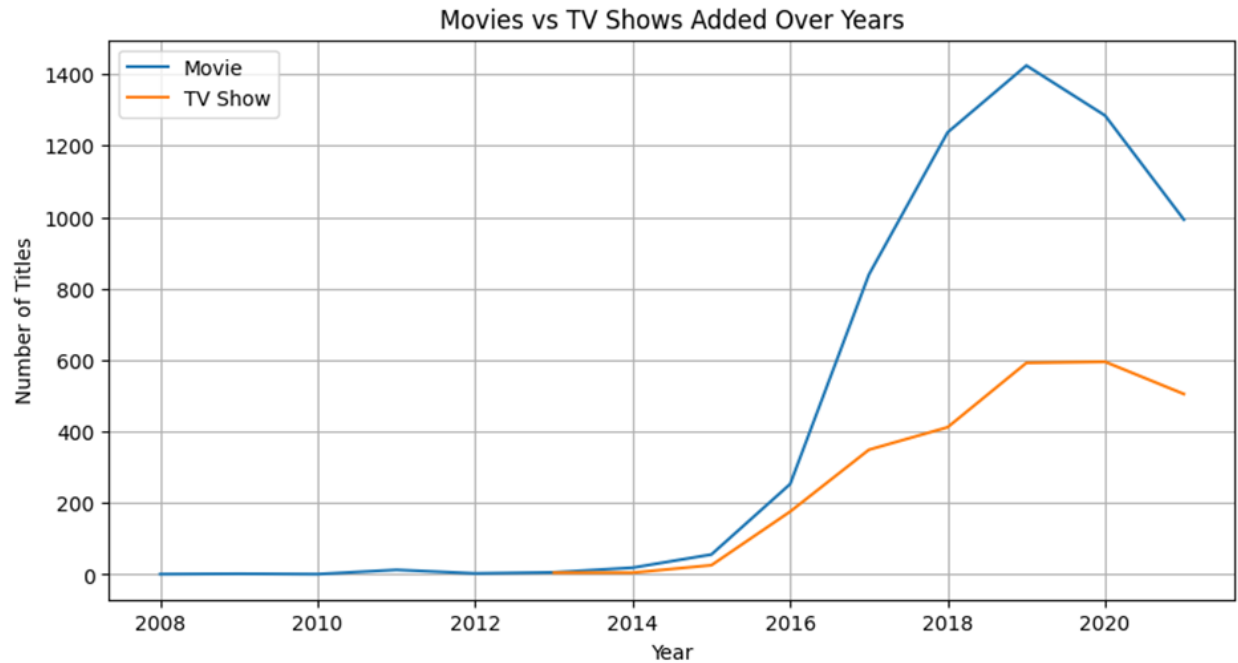


Observation:

- Content production increased rapidly in recent years.
- Netflix continuously expands its content library.
- Recent years contain the highest number of releases.
- Older releases are comparatively fewer in the dataset.

5. Movies vs TV Shows Trend Analysis

Year-wise grouped analysis was performed to compare Movies and TV Shows.

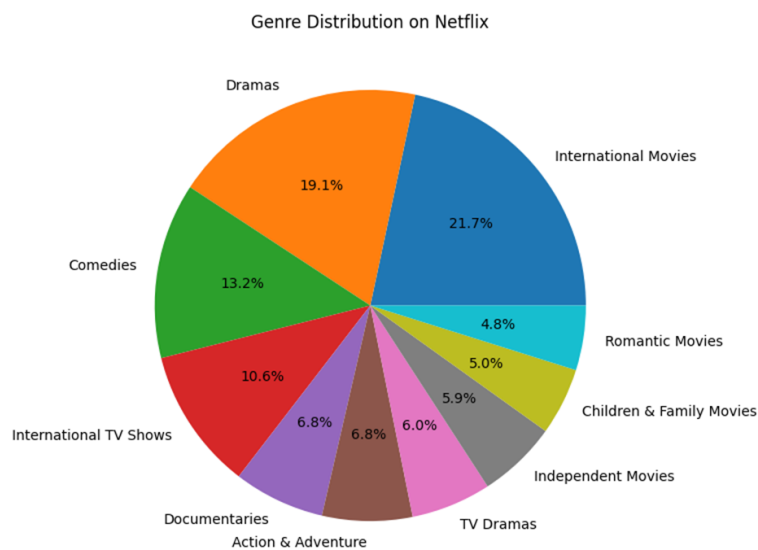


Observation:

- Movies dominate Netflix uploads across most years.
- TV Shows also increased gradually over time.
- The overall trend indicates growing demand for streaming content worldwide.

6. Genre Distribution Analysis

A pie chart was created to visualize genre popularity on Netflix.

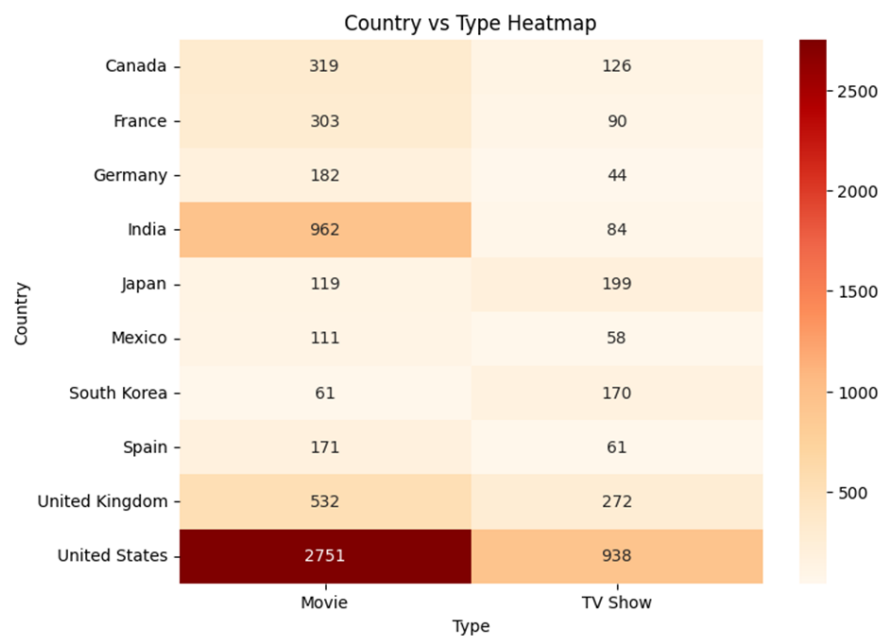


Observation:

- Certain genres are more popular than others.
 - Entertainment and drama-related genres occupy a major share.
 - Genre diversity helps attract audiences with different interests and age groups.
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7. Country vs Type Analysis

Country-wise comparison between Movies and TV Shows was performed.

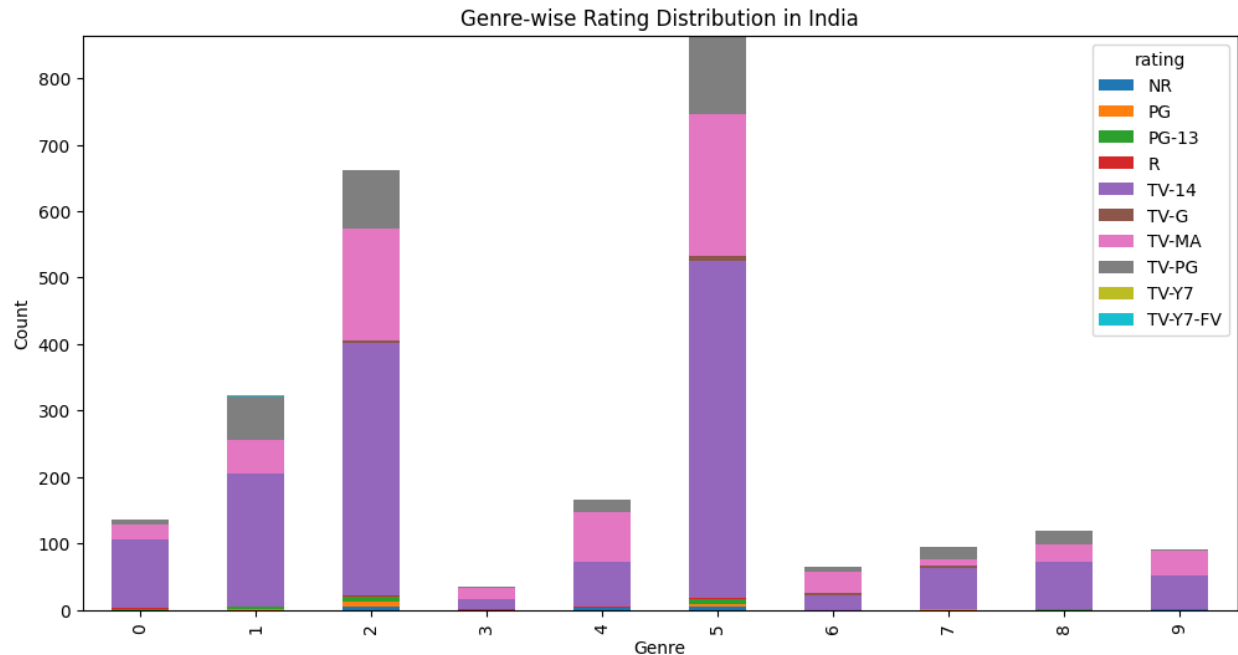


Observation:

- Some countries mainly focus on movie production.
 - Other countries contribute both Movies and TV Shows equally.
 - International content diversity strengthens Netflix's global presence.
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8. Genre-wise Rating Distribution in India

A stacked bar chart was created to analyze rating distribution across Indian content genres.



Observation:

- TV-MA has the highest count among all ratings.
- Most Netflix content targets mature audiences.
- TV-14 is the second most common rating category.
- Different genres have different rating distributions.
- Family-oriented and mature content vary significantly by genre.

Problems Faced During the Project

During the project, several challenges were encountered while performing data analysis and visualization.

Major Challenges:

1. Missing values were present in important columns such as Director, Cast, and Country.
2. Some records contained inconsistent or incomplete information.
3. Converting the `date_added` column into datetime format initially caused formatting issues.
4. Handling multiple genres and country names within a single column was difficult during analysis.
5. Choosing appropriate visualization techniques for better understanding required experimentation.
6. Some graphs produced overlapping labels and formatting problems.

7. Data preprocessing and null value handling consumed considerable time before actual analysis.
8. Managing large datasets slightly affected processing speed during exploratory analysis.
9. Exporting reports, presentations, and charts in proper format required additional formatting adjustments.

Despite these challenges, all issues were resolved successfully using data preprocessing and visualization techniques.

Key Findings

- Movies are more common than TV Shows on Netflix.
 - Netflix content has increased significantly over the years.
 - Multiple countries contribute to Netflix's global content library.
 - Genre distribution reflects audience preference diversity.
 - Mature audience content dominates the platform.
 - Data visualization helps understand streaming trends effectively.
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Conclusion

This project successfully analyzed Netflix dataset using Python data analysis libraries such as Pandas, NumPy, Matplotlib, and Seaborn.

The analysis provided valuable insights into:

- Netflix content trends
- Country-wise production
- Genre popularity
- Rating distribution
- Yearly content growth

The project demonstrates how data analytics and visualization techniques can help understand entertainment industry trends and audience behavior effectively.

Future Scope

Future improvements in this project may include:

- Development of a Netflix recommendation system.

- Sentiment analysis using audience reviews.
 - Interactive dashboard creation using Power BI or Tableau.
 - Machine learning-based content prediction models.
 - Advanced user behavior and watch-time analysis.
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Thank You